Reimagining Wait Time Through Interactive Storytelling

3B – Problem Portfolio

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Reflection on Problem Evolution

Before the start of this program, I was interested in the relationship between technology and storytelling within Disney parks. My original problem focused on the increasing use of smartphones by guests while waiting in long attraction queues. Long lines are known to be one of the least enjoyable parts of a theme park, and with modern technology many people are used to constant entertainment. As a Computer Science & Engineering major, I want to contribute to the development of new technology, without contributing to the problem of guests becoming disconnected from the environments around them.

While researching this problem, I was specifically looking at how the use of smartphones might interfere with the immersive storytelling that Disney Imagineers designed into attraction queues. Research suggests that smartphone use can reduce environmental awareness and make individuals less attentive to their surroundings (Christodoulou, A., & Roussos, P). This can lead to a lack of guest engagement with the themed environments and narratives that are specifically designed to build the story before guests even board the attraction.

My experiences throughout the program in both the US and Japan confirmed my initial assumption that many guests use their phones while waiting in line. However, it wasn't what I had expected. I saw about half of the people around me were on their phones, while the other half were interacting with family members or friends. I also noticed that some guests understood that the queue was an important part of the attraction experience. These guests spent time examining small hidden details and other storytelling elements just as the Imagineers intended. These experiences waiting in lines shifted my perspective. Instead of viewing technology as something that should be removed from the queue experience, it should be used as a tool to enhance storytelling and immersion.

Experience-Based Insights

One of my most significant realizations I had on the trip was the difference in guest behavior between the US and Japan. In the American parks, I observed many more guests using their phones in line. On the other hand, in Japan I noticed many Japanese guests spent their time talking quietly or entertaining their children without relying on devices. This contrast aligns with the Japanese culture, which has a strong emphasis on being patience. According to *JapanUp Magazine*, waiting in line in Japan is a normal social practice rooted in the cultural value of not inconveniencing others. In Japan, there was only one time when a guest was distracted by a phone and slowed down the line. In America, it happened frequently, one time involving a guest wearing headphones and reading a book in line.

Another important experience was comparing Disney mobile apps. The Tokyo Disney Resort app felt significantly more immersive using a detailed interactive 3D model of the park that guests could explore visually. Along with its technologically advanced look, it still had the same practical functions found in American Disney apps, like mobile ordering and attraction reservation systems. Seeing this interface demonstrates how computer science, animation, and user experience design can come together to support storytelling, showing how my academic background can contribute directly to guest immersion.

Stakeholder & Ecological Reassessment

These observations led me to reconsider the benefits of my proposed solution from a stakeholder and ecological perspective. The way the Disney apps are designed now, guests are frequently using them, but most interactions are brief. Introducing an interactive queue experience directly into the park apps could increase engagement while also improving the guest

experience. It would bring in longer and more meaningful interactions within that app that benefit both the guests with a more immersive experience, and Disney stakeholders with increased usage of its digital platform.

From an ecological perspective, adding these digital features may offer a more sustainable solution. Disney can enhance existing spaces through mobile experiences, rather than investing in large scale construction projects to make older queues more physically interactive. For example, recent refurbishment work done at The Twilight Zone Tower of Terror proves just how much effort is often required to renovate queue environments (WDW News Today). Digital applications in queues could supplement existing environments without the environmental impact that comes with large renovation projects. New insight I gained from the trip is that technology does not always compete with immersion.

Refined Solutions

In my original proposal, I thought that Disney needed to create an entirely new system for these technological experiences in the queue. I quickly discovered that Disney already offers a similar concept through the Play Disney Parks app, an app designed to "turn wait time into play time" by giving guests attraction themed games and activities while waiting in line (Contributor, M. L. D. N). Peter Pan's Flight queue in Disneyland stood out to me. Guests search for hidden symbols carved into the wood and check them off to unlock an interactive surprise with Tinkerbell. I soon noticed a few flaws: guests could simply click every symbol without participating in the scavenger hunt, and not many people knew about the Disney app. This led me to refine my solution by integrating these games directly into the Disney Park app. This would significantly increase guest participation and visibility. I would also like to expand the use of augmented reality features, in the Peter Pan queue guests could scan the physical carvings of

the symbols to verify their progress in the scavenger hunt. This would encourage actual participation in the games and make rewards feel meaningful.

A similar solution I thought of was for the Toy Story Mania attraction, a queue that was scattered with themed posters. These posters could be integrated with image recognition technology to bring the posters to life with animated 3D visuals on the guests' phones. This way, the app combines pre-existing physical storytelling with digital interactions that allow guests to engage with the environment. Overall, my refined solution would emphasize creativity, sustainability, and immersive design by using technology as an enhancement to the guest experience, not a distraction.

Personal & Professional Growth

My experiences throughout this program influenced both my academic interests and career goals. Before this program, I had never fully understood the role software development plays in shaping guest experiences. After seeing how heavily guests rely on the Disney apps throughout their visits, I have a new appreciation for the impact software engineers have on operations within the parks. This program helped me grow my understanding of all the types of Imagineering. Disney reports that there are about 1,500 Imagineers working across roughly 140 different disciplines (KORN, M). This shows how there is such a wide range of expertise required to create a Disney experience. It broadened my perspective on all the potential career paths within the entertainment industry, whether it's through mobile app development, production technology, user experience design, or animation. This trip has led me to realize that whether through Disney or another entertainment company, I hope to connect technology and creativity in ways that enhance how people engage with storytelling.

References

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